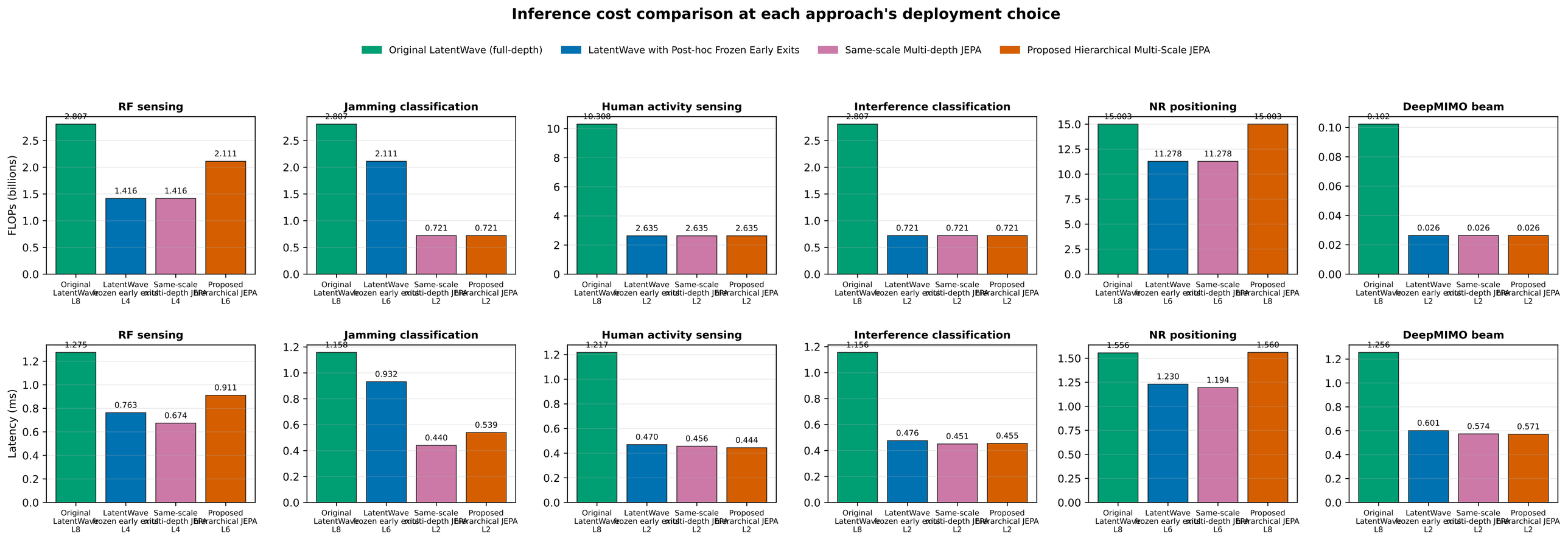
Inference cost comparison at each approach's deployment choice
Original LatentWave (full-depth)
LatentWave with Post-hoc Frozen Early Exits
Same-scale Multi-depth JEPA
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Jamming classification
Human activity sensing
Interference classification
NR positioning
DeepMIMO beam
FLOPs (billions)
Latency (ms)
2.807
1.416
1.416
2.111
2.807
2.111
0.721
0.721
10.308
2.635
2.635
2.635
2.807
0.721
0.721
0.721
15.003
11.278
11.278
15.003
0.102
0.026
0.026
0.026
1.275
0.763
0.674
0.911
1.158
0.932
0.440
0.539
1.217
0.470
0.456
0.444
1.156
0.476
0.451
0.455
1.556
1.230
1.194
1.560
1.256
0.601
0.574
0.571
Original LatentWave L8
LatentWave with frozen early exits L4
LatentWave with frozen early exits L6
LatentWave with frozen early exits L2
Same-scale multi-depth JEPA L4
Same-scale multi-depth JEPA L6
Same-scale multi-depth JEPA L2
Proposed Hierarchical JEPA L6
Proposed Hierarchical JEPA L8
Proposed Hierarchical JEPA L2
Batch-1 synchronized CUDA timing on the RTX 5070 Ti; lower is better. Layer labels are printed below each method.